Synaptic effects of interleukin-6 on human iPSC-derived dopaminergic neurons

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What transcriptomic changes are induced by IL-6 treatment in iPSC-derived cells, and are these effects consistent across sex and cell line?

Mary Lopez

Experimental design

- Cells are sourced from skin biopsy or blood sample from health donors that do not identify as depressed. The human induced pluripotent stem cells (iPSCs) were generated from these somatic cells. They were then differentiated into DA neuronal cells.

- IL-6 administration decreases availability and release of DA in the ventral striatum

System: iPSC-derived dopaminergic neurons

Treatment: IL-6 vs Control

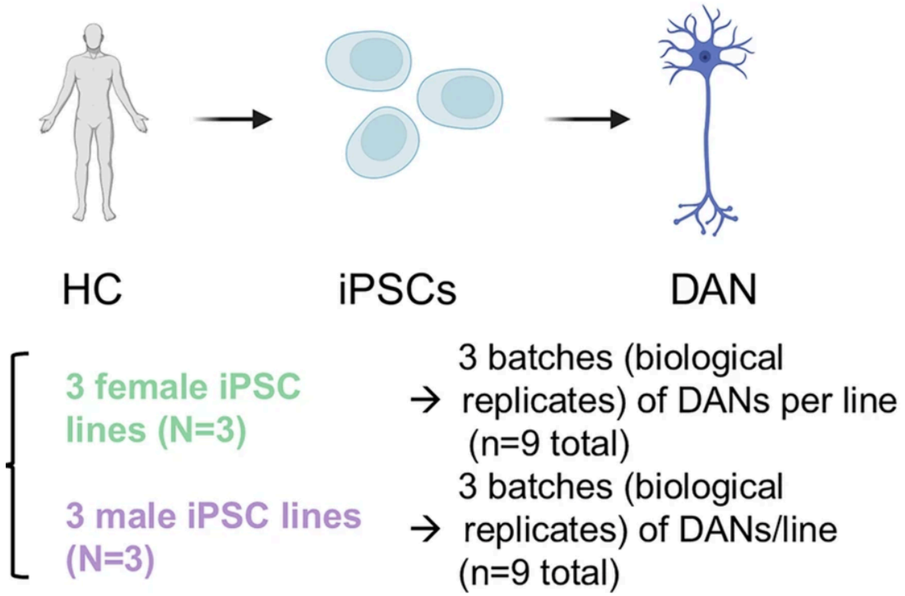
Variables: Sex, Cell Line

Replicates: 3 biological replicates per condition

3 technical replicates per line

Outcome: Gene expression (RNA-seq)

Comparison: Treated vs Control, M vs F



PIPELINE

GEO Data (GSE296202)

↓

Preprocessing

- Extract counts
- Map Entrez → Gene symbols

↓

Filtering

- Remove low expression genes (CPM filter)

↓

Normalization

- TMM (edgeR)

↓

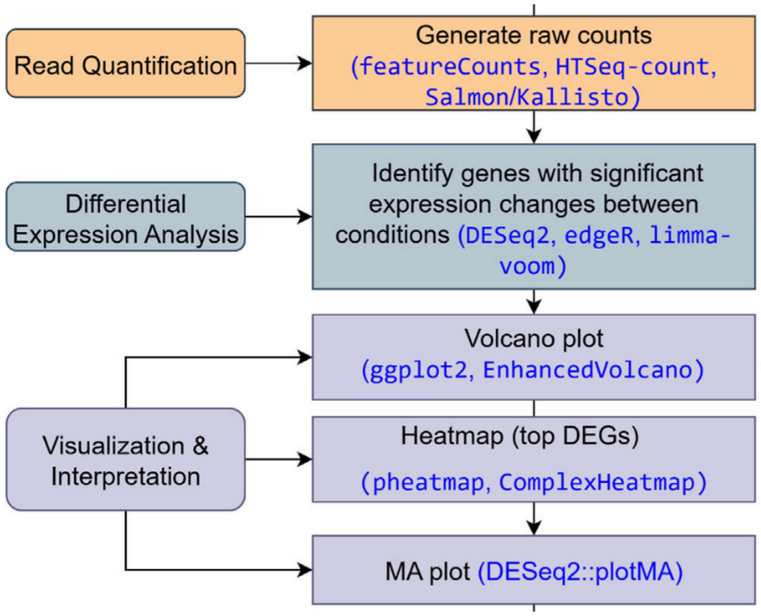
Transformation

- voom (mean-variance modeling)

↓

Statistical Modeling

- Design: ~ Sex + Line + Treatment
- limma (linear model + eBayes)



Differential Expression

- logFC, p-value, FDR

↓

Gene Ranking

↓

Enrichment Analysis

- fgsea / gProfiler (GO, REAC)

↓

Cytoscape

- Import DE genes/pathways
- Build network (STRING)
- Visualize clusters

Filtering and normalization

```
keep_all <- rowSums(cpm(y_all) > 1) >= 4
keep_f <- rowSums(cpm(y_f) > 1) >= 3

y_f <- calcNormFactors(y_f, method = “TMM")
```

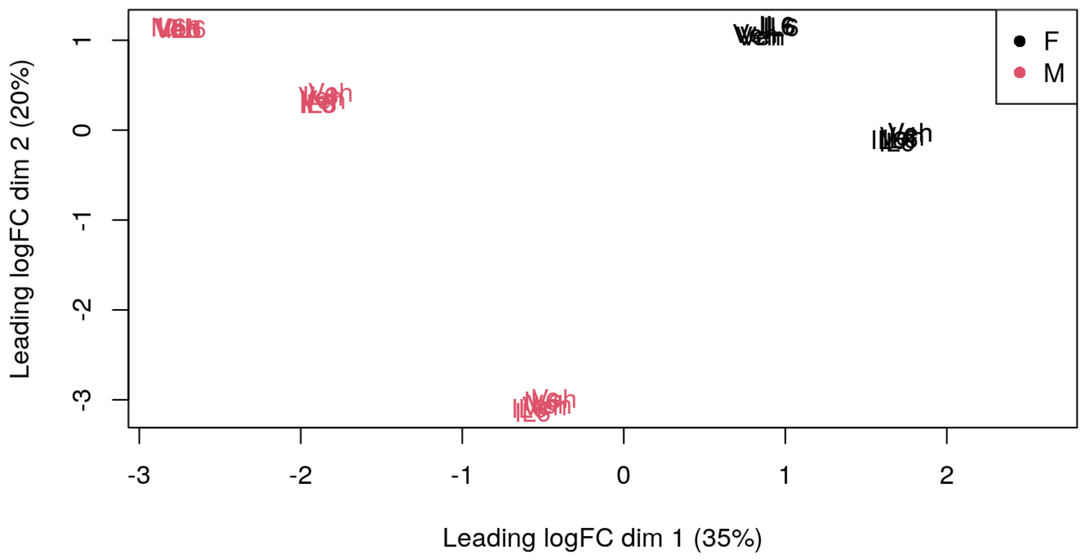
```
design_all <- model.matrix(~ Sex + Line + Treatment, data = pheno_bio)
design_f <- model.matrix(~ Line + Treatment, data = pheno_f)
```

```
All-samples: raw p < 0.05 = 7

Female samples
Raw p < 0.05: 452
FDR < 0.05: 0
Up-regulated genes for ORA: 110
Down-regulated genes for ORA: 122
```

```
Collapse technical replicates into biological samples
base_names <- sub("\\.[ABC]$", "", colnames(counts_hgnc))

counts_bio <- sapply(unique(base_names), function(x) {
  rowSums(counts_hgnc[, base_names == x, drop = FALSE])
}
counts_bio <- as.matrix(counts_bio)
mode(counts_bio) <- “numeric"
```

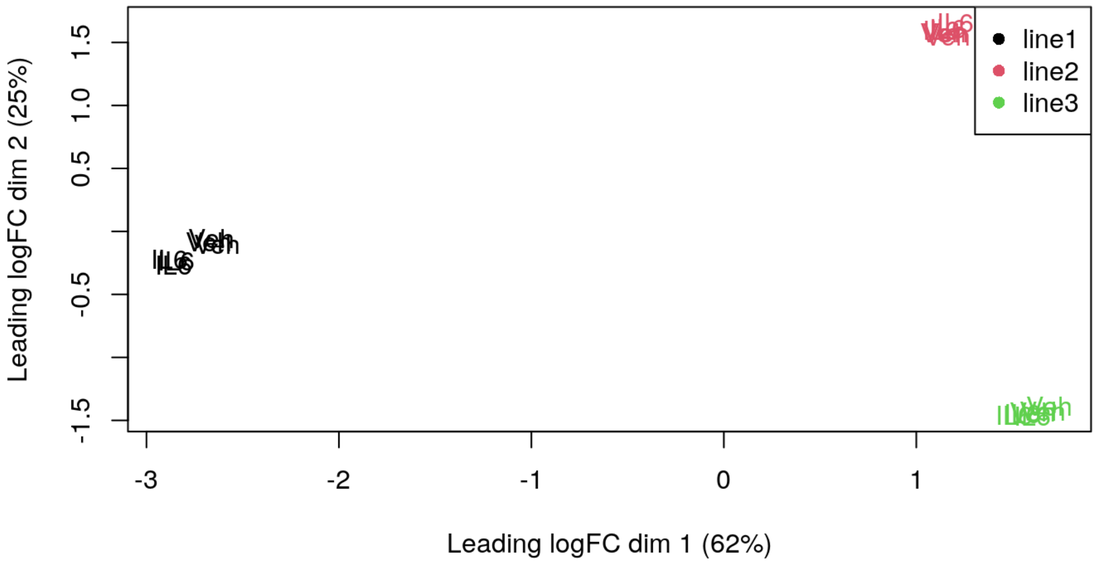
Methods	Sequencing depth correction	Gene length correction	Library composition correction	Suitable for DE analysis	Notes
CPM (Counts per Million)	Yes	No	No	No	Simple scaling by total reads; affected by highly expressed genes
RPKM / FPKM (Reads/ fragments per Kilobase of Transcript, per Million Mapped Reads)	Yes	Yes	No	No	Adjusts for gene length; still affected by library composition
TPM (Transcripts per Million)	Yes	Yes	Partial	No	Scales sample to constant total (1M), reducing composition bias; good for visualization and cross-sample comparison
median-of-ratios	Yes	No	Yes	Yes	Implemented in DESeq2; affected by expression shifts
TMM (Trimmed Mean of M-values)	Yes	No	Yes	Yes	Implemented in edgeR; affected by over-trimming genes

MDS PLOT: ALL SAMPLES



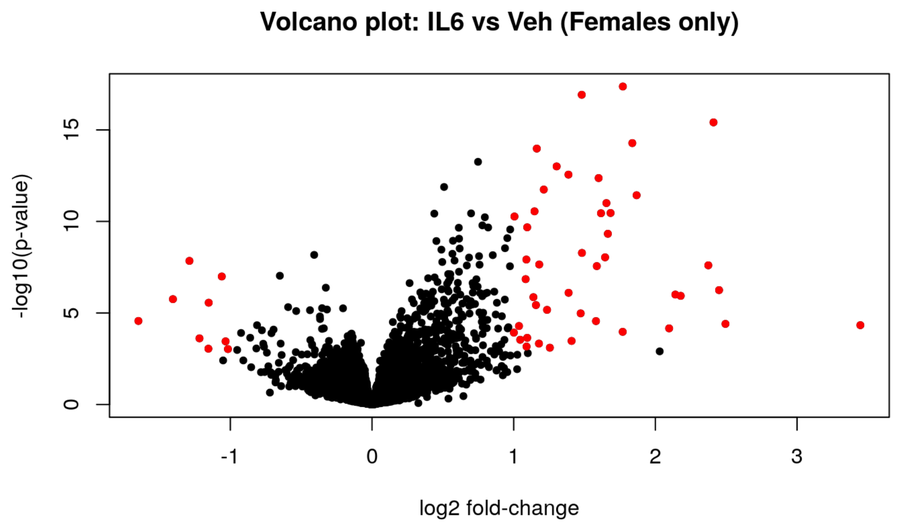

```r
# MDS plot for ALL samples
logCPM_all <- cpm(y_all, log = TRUE, prior.count = 1)
plotMDS(
  logCPM_all,
  labels = pheno$Treatment,
  col    = as.integer(pheno$Sex)
)
legend("topright",
       legend = levels(pheno$Sex),
       col    = 1:2,
       pch    = 16)
```

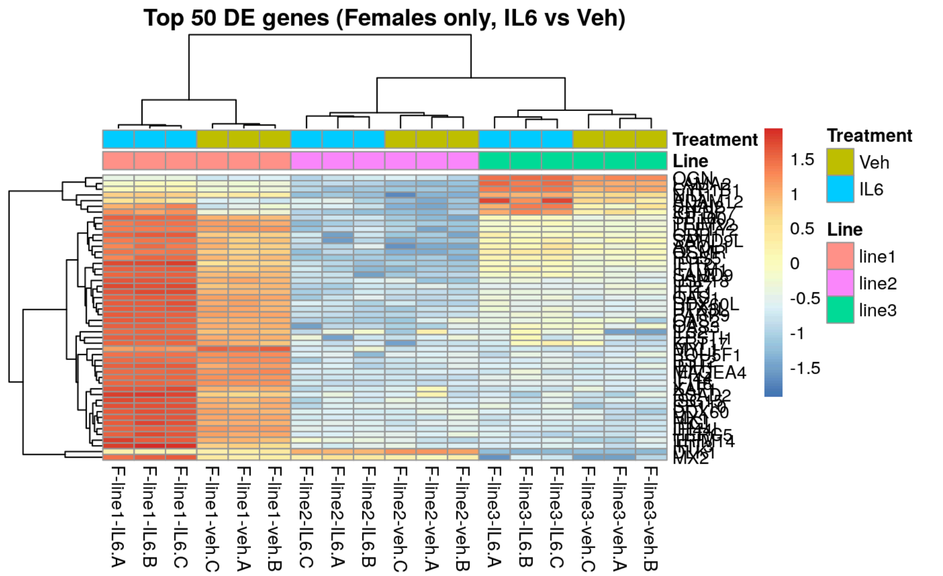
FEMALE SUBSET




```r
logCPM_f <- cpm(y_f, log = TRUE, prior.count = 1)
plotMDS(
  logCPM_f,
  labels = pheno_f$Treatment,
  col    = as.integer(pheno_f$Line)
)
legend("topright",
       legend = levels(pheno_f$Line),
       col    = 1:length(levels(pheno_f$Line)),
       pch    = 16)
```

Differential Expression plot





The volcano plot identifies significantly differentially expressed genes, showing a directional upregulation under IL-6.

There is many more upregulated differentially expressed genes than down regulated differentially expressed genes

This heat map is difficult to analyze the differentially expressed genes cluster more around cell line than anything else however within each cell line there is marginal difference between treatment and control.

This suggests that while IL-6 induces differential expression, cell line acts as a major source of variability, reinforcing the need to account for it in the linear model

Top female-only differential expression results ranked by nominal p-value.

gene	logFC	P.Value	adj.P.Val
SP100	1.1361463	0.0000104	0.1347349
NID1	0.5066807	0.0000179	0.1347349
IFITM1	1.2646945	0.0000270	0.1356471
SAMD9	1.3330972	0.0000605	0.2276387
LAMA2	0.4313861	0.0000814	0.2449237
DDX60L	1.1571138	0.0001500	0.3335515
ADAM12	0.4889619	0.0001655	0.3335515
IFI44	0.6970068	0.0001774	0.3335515
OGN	0.5914880	0.0002301	0.3370826
TRIM14	0.9251564	0.0002599	0.3370826
RGS5	0.6045751	0.0002955	0.3370826
GBP1	1.5363687	0.0002996	0.3370826
ISG15	1.5079413	0.0003039	0.3370826
CYP1B1	0.4448785	0.0003137	0.3370826
TRIM22	0.5887926	0.0003386	0.3395712

Thresholded gene set enrichment analysis Over representation analysis using gprofiler

Upregulated response

```
sources = c("GO:BP")
```

Table 1. Top 10 enriched GO Biological Process terms from the up-regulated female-only gene set.

term_name	p_value	intersection_size	term_size
defense response to virus	0	28	319
response to virus	0	30	414
biological process involved in interspecies interaction between organisms	0	47	1759
response to external biotic stimulus	0	45	1595
response to other organism	0	45	1592
response to biotic stimulus	0	45	1634
response to external stimulus	0	53	2415
innate immune response	0	36	984
defense response to other organism	0	39	1263
defense response to symbiont	0	36	1082

```
sources = c("GO:BP", “REAC”)
```

Table 1. Top 10 enriched GO Biological Process terms from the up-regulated female-only gene set.

term_name	p_value	intersection_size	term_size
Interferon Signaling	0.0000000	22	263
Interferon alpha/beta signaling	0.0000000	13	76
Cytokine Signaling in Immune system	0.0000000	26	780
Immune System	0.0000029	37	2085
SARS-CoV-1 Infection	0.0000063	10	140
Negative regulators of DDX58/IFIH1 signaling	0.0000142	6	35
Interferon gamma signaling	0.0000207	8	91
SARS-CoV-1 activates/modulates innate immune responses	0.0000245	6	40
DDX58/IFIH1-mediated induction of interferon-alpha/beta	0.0001141	7	82
SARS-CoV-1-host interactions	0.0002767	7	95

Down regulated response

```
gost_down <- gost(
  query = down_genes,
  organism = "hsapiens",
  correction_method = "fdr",
  sources = c("GO:BP", "GO:MF",
"GO:CC")
)
```

several down-regulated pathways were related to muscle function and ion channel activity, including muscle system process, striated muscle contraction, skeletal muscle contraction, and inward rectifier potassium channel complex. Although these pathways are not specific to dopaminergic neurons, they may reflect broader alterations in cellular excitability and functional output. These findings are consistent with literature linking inflammation to motor related symptoms in depression. The observed down-regulation of muscle and contraction related processes may reflect transcriptional changes associated with impaired neuronal activity and motor-related function in the context of inflammation.

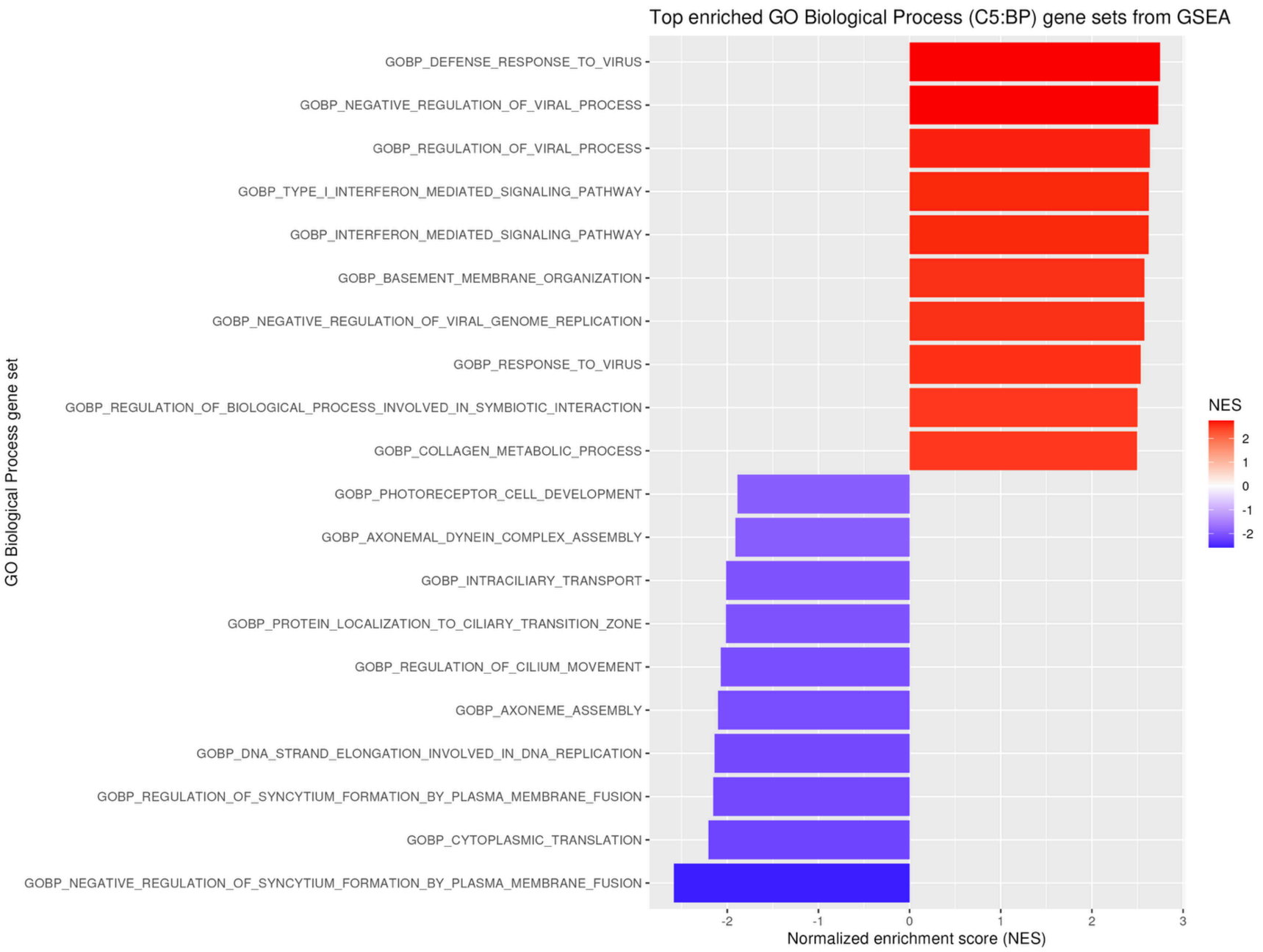
Table 2. Top 10 enriched GO Biological Process terms from the down-regulated female-only gene set.

term_name	p_value	intersection_size	term_size
muscle system process	0.0161997	10	438
striated muscle contraction	0.0161997	7	189
regulation of skeletal muscle adaptation	0.0161997	3	12
regulation of muscle adaptation	0.0161997	5	79
skeletal muscle contraction	0.0255837	4	47
inward rectifier potassium channel complex	0.0390718	2	4
extracellular region	0.0404091	33	4258
collagen trimer	0.0404091	4	91
extracellular space	0.0404091	28	3315
endoplasmic reticulum	0.0404091	21	2095

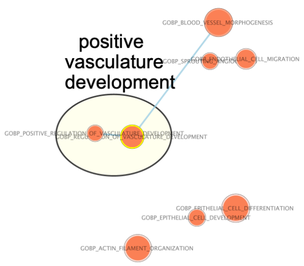
FGSEA PLOT

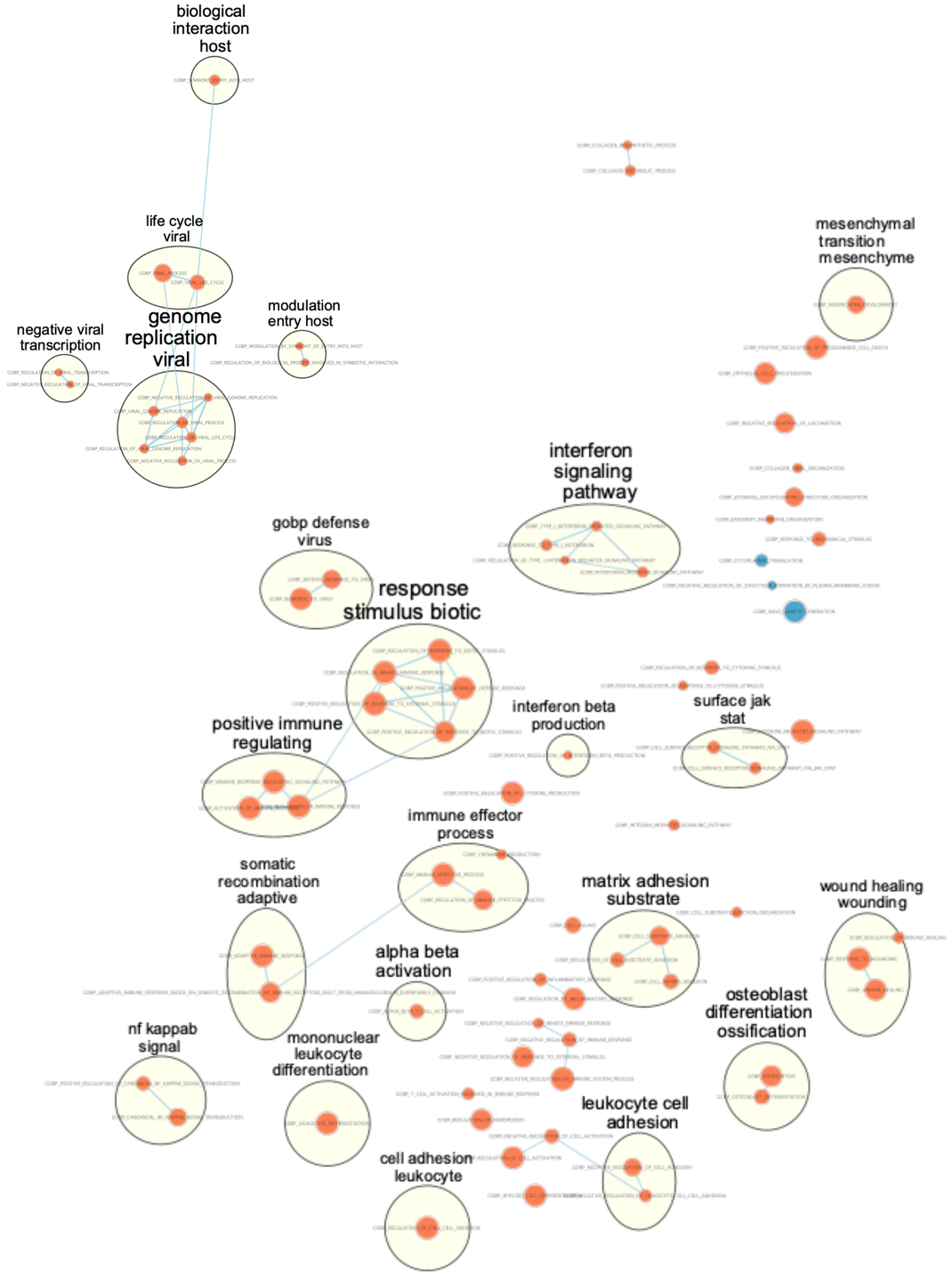
```
msig_h <- msigdbr(
  species = "Homo sapiens",
  collection = "C5",
  subcollection = "BP"
)
pathways <-
split(msig_h$gene_symbol,
msig_h$gs_name)

fgsea_res <- fgseaMultilevel(
  pathways = pathways,
  stats = gene_ranks,
  minSize = 15,
  maxSize = 500
)
```

```
Significant GO Biological
Process (C5:BP) gene sets from
GSEA: 705
Positively enriched gene sets
(NES > 0): 659
Negatively enriched gene sets
(NES < 0): 46
```

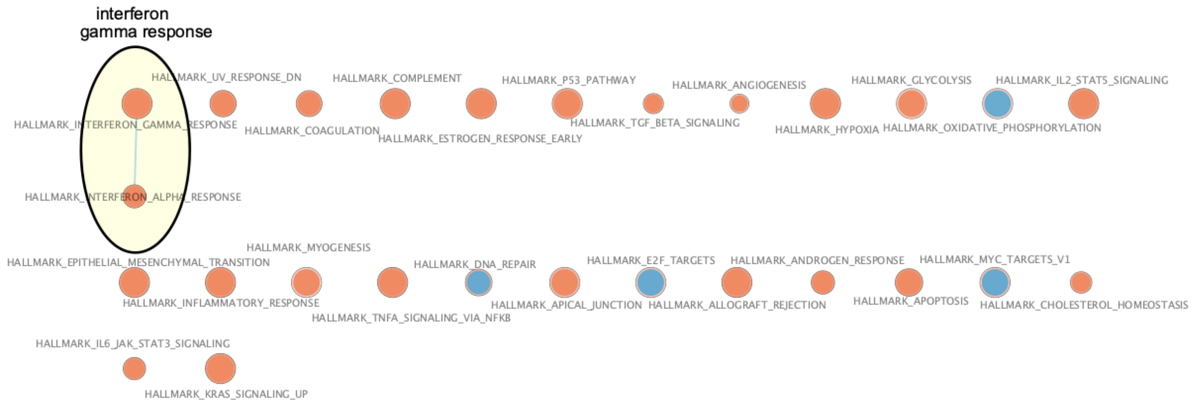


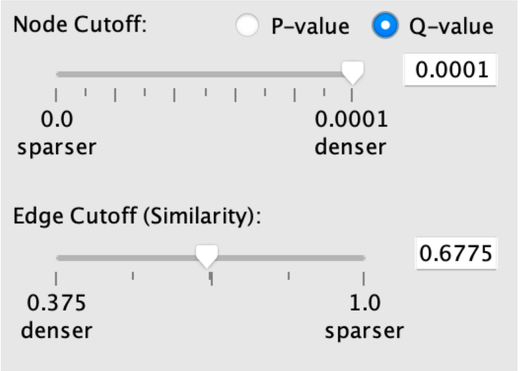
Cytoscape





Hallmark:

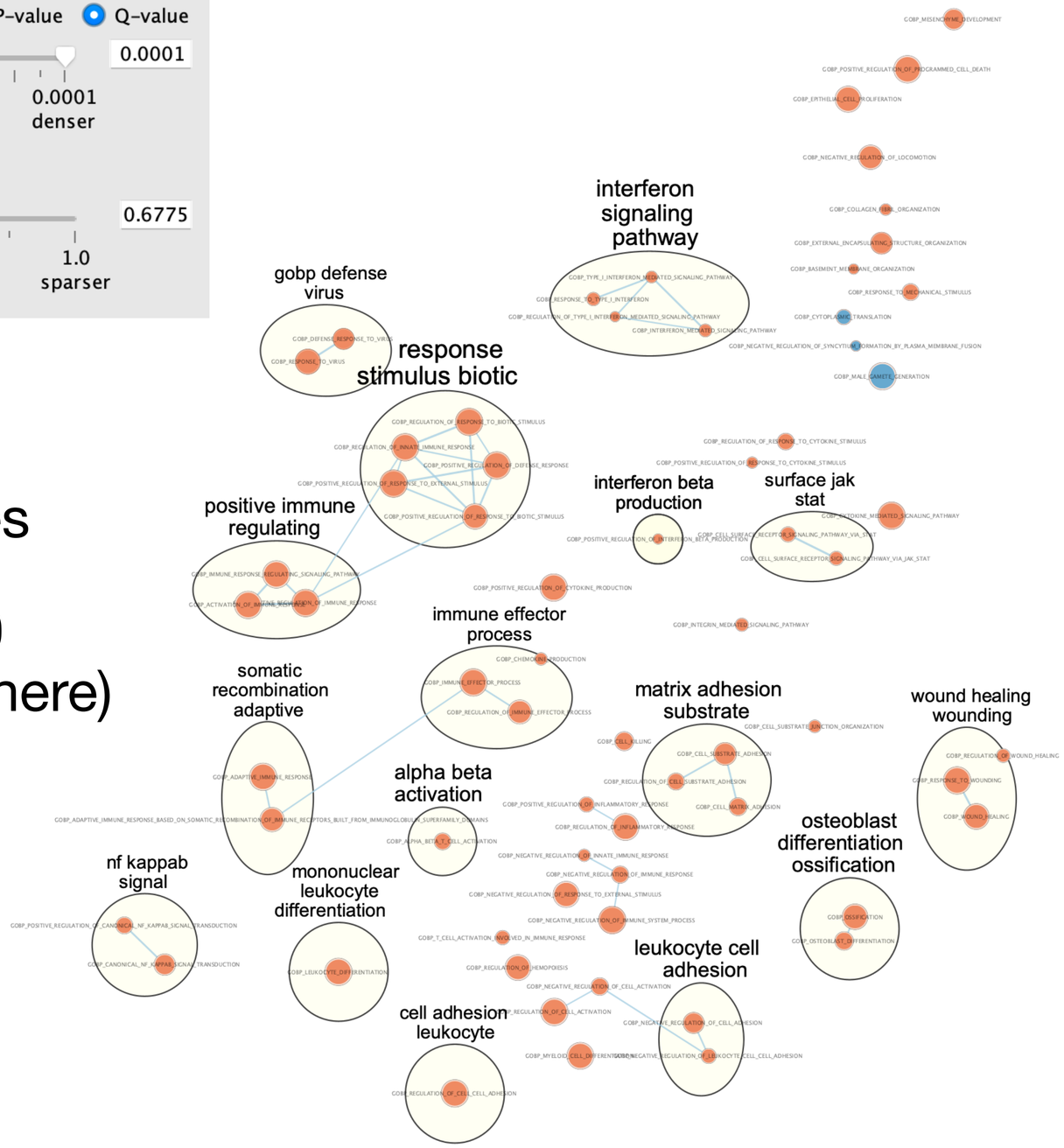




98 nodes

242 edges

(about 30 pictured here)



- Mechanistically, IL-6 signaling is known to act through the JAK/STAT pathway, leading to activation of STAT transcription factors and downstream induction of inflammatory and immune-related genes (Hu et al., 2021).
- The presence of JAK/STAT-related and immune effector pathways in both ORA and GSEA results further supports the conclusion that IL-6 drives a coordinated inflammatory transcriptional program in these cells
- while the interferon and antiviral clusters indicate downstream activation of immune transcriptional programs. The identification of immune effector, regulatory, and wound healing pathways suggests that IL-6 induces not only signaling responses but also broader functional changes in cellular behavior.related genes: IFITM1, IFI44, ISG15, GBP1

References

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